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Perturb-seq: High throughput functional genomics at single cell resolution



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In two papers published today in *Cell*, two teams working collaboratively from UCSF and the Broad Institute describe Perturb-seq, a novel approach to high-throughput functional genomics enabled by the combination of single cell RNA-seq with CRISPR perturbations. Both teams took advantage of the massive throughput of the 10x Chromium™ Single Cell 3' Solution to screen and sequence the CRISPR-perturbed cells on an unprecedented scale.

The UCSF-led team introduced a library of barcoded CRISPR interference (CRISPRi) vectors targeting ~100 different genes into a population of cells, such that each cell received only a single CRISPRi vector. They then used the 10x Chromium™ Single Cell 3' Solution to identify both the introduced CRISPRi vector and its effect on gene expression in each cell. To demonstrate the power of this approach, [Adamson and Norman *et al*](#) dissected the unfolded protein response (UPR), a complex cellular stress-coping pathway that is frequently found to be mis-regulated in neurodegenerative disease. While, the Broad-led team of [Dixit and Parnas *et al*](#) used CRISPR/Cas9 nucleases to cut DNA and inactivate genes for transcription factor proteins (TFs) to study the effect of TFs on genes, programs, and states in LPS response in immune cells.

The use of CRISPR/Cas9 technology to alter gene expression on a genome-wide is a powerful tool for studying gene function and engineering cellular systems^{1,2}. The introduction of Perturb-seq enables CRISPR-based experiments at an



unprecedented scale with major improvements over traditional high-throughput screening methods:

- **Experimental scalability**- Profiling gene expression and CRISPR barcodes at the single cell level allows researchers to test the function of hundreds and potentially thousands of genes in a single experiment, instead of painstakingly perturbing them one at a time.
- **Cost and time savings** - Using unbiased gene expression profiling as a general read-out of changes in cellular state allows researchers to skip the costly and lengthy development of traditional reporter assays.

Read more about these publications in the UCSF [Press Release](#) and visit our website to learn more about the [Chromium™ Single Cell 3' Solution](#).

Watch Britt Adamson's talk, "[Perturb-seq: Single-cell CRISPRi screens interrogate the Unfolded Protein Response](#)", from our 2017 Bay Area User Group Meeting.

References

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2. Ophir Shalem, Neville E. Sanjana, Ella Hartenian, Xi Shi, David A. Scott, Tarjei S. Mikkelsen, Dirk Heckl, Benjamin L. Ebert, David E. Root, John G. Doench, Feng Zhang Genome-Scale CRISPR-Cas9 Knockout Screening in Human Cells *Science* (2014) Vol. 343, Issue 6166, pp. 84-87 DOI: [10.1126/science.1247005](https://doi.org/10.1126/science.1247005)
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